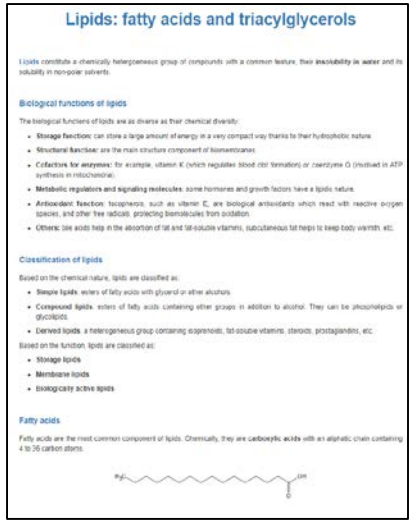
UNIT 2.
Lecture 2.7. Lipids


Lipids: fatty acids and triacylglycerols

Lipids constitute a chemically heterogeneous group of compounds with a common feature, their **insolubility in water** and its solubility in non-polar solvents.

Biological functions of lipids

The biological functions of lipids are as diverse as their chemical diversity:

- **Storage function:** can store a large amount of energy in a very compact way thanks to their hydrophobic nature.
- **Structural function:** are the main structure component of biomembranes.
- **Cofactors for enzymes:** for example, vitamin K (which regulates blood clot formation) or coenzyme Q (involved in ATP synthesis in mitochondria).
- **Metabolic regulators and signaling molecules:** some hormones and growth factors have a lipidic nature.
- **Antioxidant function:** tocopherols, such as vitamin E, are biological antioxidants which react with reactive oxygen species, and other free radicals, protecting biomolecules from oxidation.
- **Others:** bile acids help in the absorption of fat and fat-soluble vitamins, subcutaneous fat helps to keep body warmth, etc.

Classification of lipids

Based on the chemical nature, lipids are classified as:

- **Simple lipids:** esters of fatty acids with glycerol or other alcohols.
- **Compound lipids:** esters of fatty acids containing other groups in addition to alcohol. They can be phospholipids or glycolipids.
- **Derived lipids:** a heterogeneous group containing isoprenoids, fat-soluble vitamins, steroids, prostaglandins, etc.

Based on the function, lipids are classified as:

- **Storage lipids**
- **Membrane lipids**
- **Biologically active lipids**

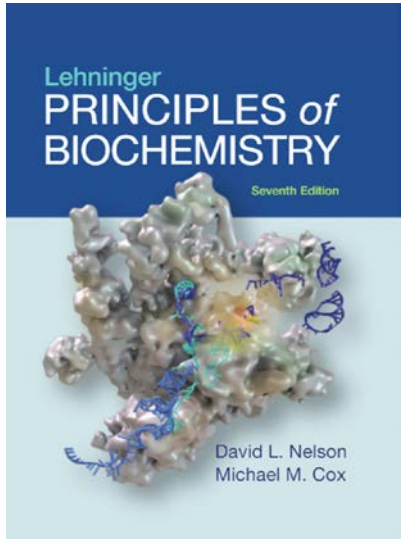
Fatty acids

Fatty acids are the most common component of lipids. Chemically, they are **carboxylic acids** with an aliphatic chain containing 4 to 36 carbon atoms.

Study material

ADI document:

Lipids: fatty acids and TAG



Reading

Lehninger's PoB Chapter 10:

10.1. Storage lipids

10.3. Biologically active lipids

UNIT 2.
Lecture 2.7. Lipids

Learning goals:

- Biological roles of lipids
- Classification of lipids
- Structure and properties of fatty acids
- Structure and properties of storage lipids
- Structure and properties of regulatory lipids
- *In Unit 3, Lec 3.1. and Lec 3.2. you will study the structure and properties of membrane lipids*

Lipids are a chemically diverse group of compounds with one characteristic in common:

Their low solubility in water



Biological functions of lipids

- Storage function
  - high energy nutrients
  - good packing properties
- Structural function
  - main structure of cell membranes
- Cofactors for enzymes
  - vitamin K: blood clot formation
  - coenzyme Q: ATP synthesis in mitochondria
- Signaling molecules
  - paracrine hormones (act locally)
  - steroid hormones (act body-wide)
  - growth factors
  - vitamins A and D (hormone precursors)
- Antioxidants
  - vitamin E

Classification of lipids

- Lipids that contain fatty acids:
  - can be further separated into:
    - storage lipids
    - membrane lipids
- Biologically active lipids that do not contain fatty acids:
  - cholesterol, vitamins, pigments, etc.

Lipids that contain fatty acids

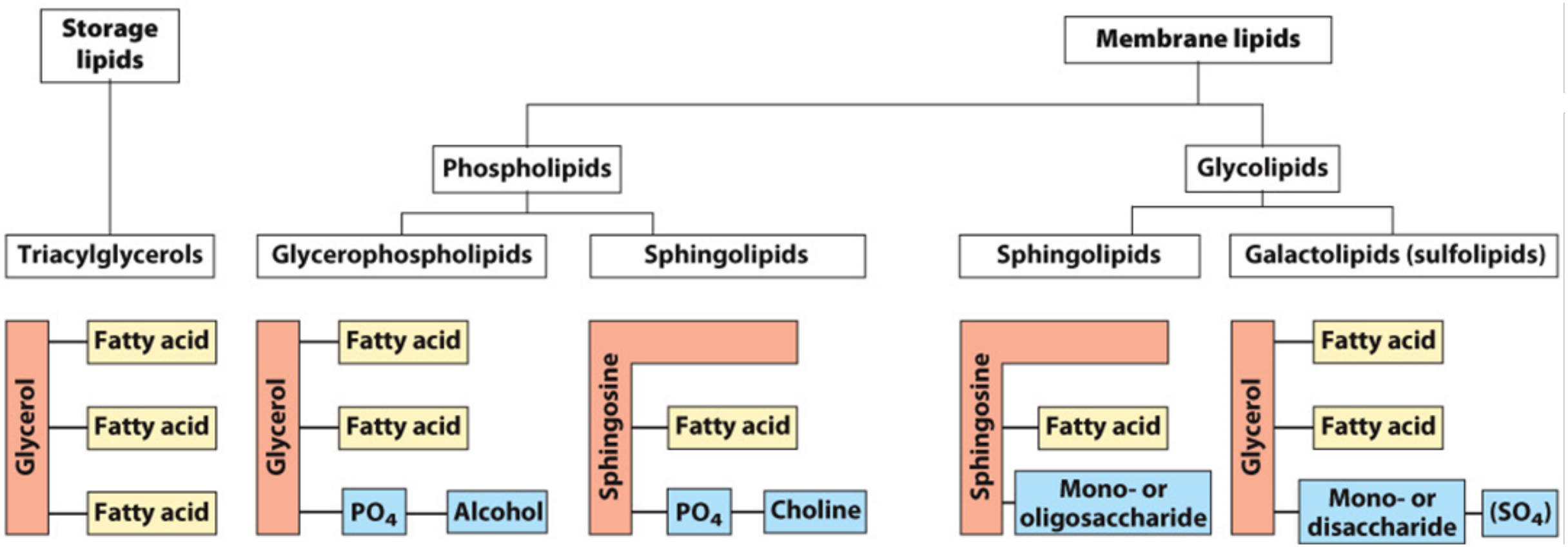


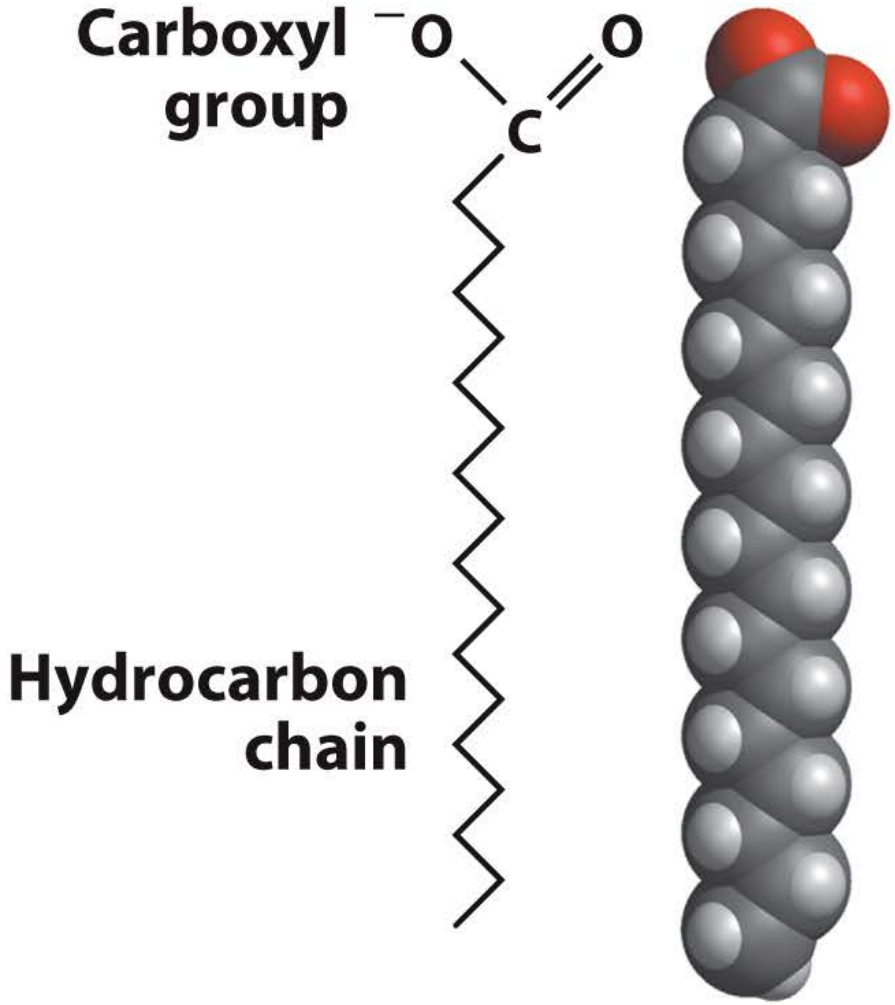
Figure 10-6
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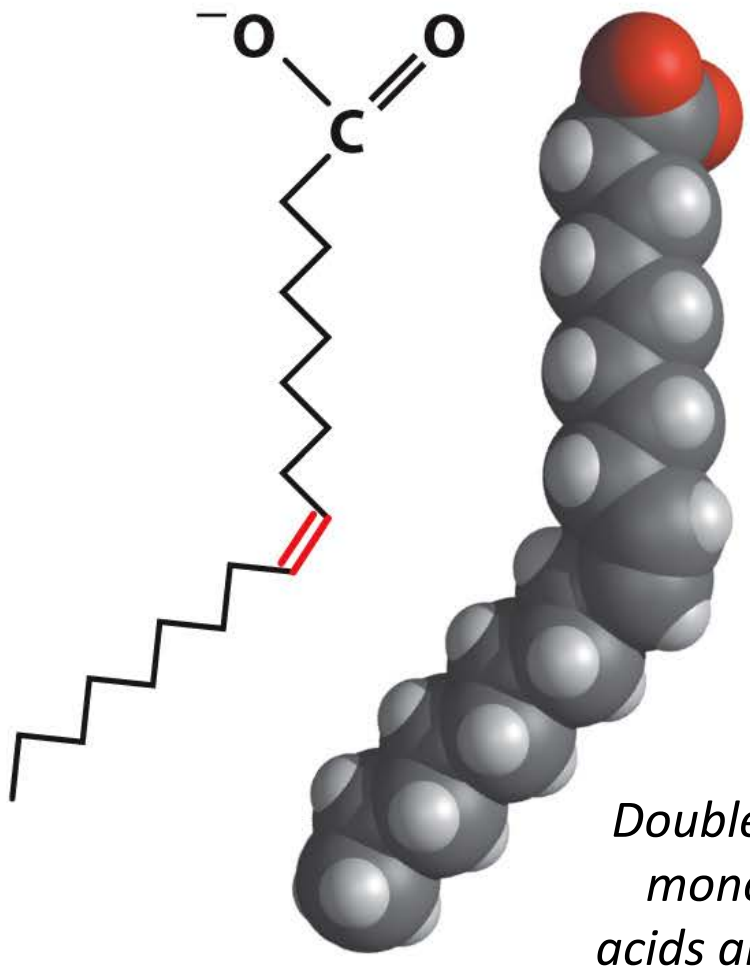
Fatty acids

O

- Carboxylic acids with hydrocarbon chains containing between 4 to 36 carbons
  - Almost all natural fatty acids have an even number of carbons
  - Most natural fatty acids are unbranched
- Saturated: no double bonds between carbons in the alkyl chain
- Monounsaturated: one double bond between carbons in the alkyl chain
- Polyunsaturated (PUFA): two or more double bonds in the alkyl chain

Fatty acids

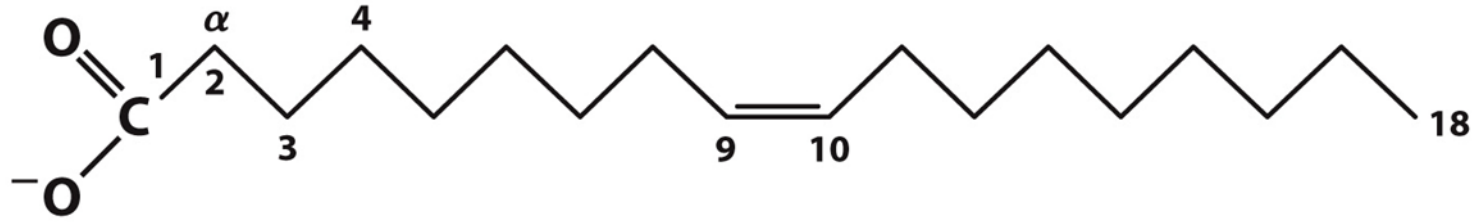




Double bonds in natural monosaturated fatty acids are commonly in cis configuration

Figure 10-1ab
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Fatty acid nomenclature



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- systematic name: *cis*-9-octadecenoic acid
- common name: oleic acid
- delta numbering of carbon skeleton: $18{:}1^{\Delta 9}$
  - describes location of the first carbon of the alkene in relationship to the carbonyl carbon
- omega numbering of carbon skeleton: $18{:}1^{\omega 9}$
  - describes location of the first carbon of the alkene in relationship to the terminal methyl

Saturated fatty acids

TABLE 10-1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

Carbon skeleton	Structure[a]	Systematic name[b]	Common name (derivation)	Melting point (˚C)	Solubility at 30 ˚C (mg/g solvent) Water	Benzene
12:0	$CH_3(CH_2)_{10}COOH$	*n*-Dodecanoic acid	Lauric acid (Latin *laurus*, "laurel plant")	44.2	0.063	2,600
14:0	$CH_3(CH_2)_{12}COOH$	*n*-Tetradecanoic acid	Myristic acid (Latin *Myristica*, nutmeg genus)	53.9	0.024	874
16:0	$CH_3(CH_2)_{14}COOH$	*n*-Hexadecanoic acid	Palmitic acid (Latin *palma* "palm tree")	63.1	0.0083	348
18:0	$CH_3(CH_2)_{16}COOH$	*n*-Octadecanoic acid	Stearic acid (Greek *stear*, "hard fat")	69.6	0.0034	124
20:0	$CH_3(CH_2)_{18}COOH$	*n*-Eicosanoic acid	Arachidic acid (Latin *Arachis*, legume genus)	76.5		
24:0	$CH_3(CH_2)_{22}COOH$	*n*-Tetracosanoic acid	Lignoceric acid (Latin *lignum*, "wood" + *cera*, "wax")	86.0		

[a]All acids are shown in their nonionized form. At pH 7, all free fatty acids have an ionized carboxylate. Note that numbering of carbon atoms begins at the carboxyl carbon.
[b]The prefix *n*- indicates the "normal" unbranched structure. For instance, "dodecanoic" simply indicates 12 carbon atoms, which could be arranged in a variety of branched forms; "*n*-dodecanic" specifies the linear, unbranched form. For unsaturated fatty acids, the configuration of each double bond is indicated; in biological fatty acids the configuration is almost always cis.

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Unsaturated fatty acids

TABLE 10-1 Some Naturally Occurring Fatty Acids: Structure, Properties, and Nomenclature

Carbon skeleton	Structure[a]	Systematic name[b]	Common name (derivation)	Melting point (˚C)	Solubility at 30 ˚C (mg/g solvent) Water	Benzene
16:1(Δ^9)	$CH_3(CH_2)_5CH=CH(CH_2)_7COOH$	*cis*-9-Hexadecenoic acid	Palmitoleic acid	1 to –0.5		
18:1(Δ^9)	$CH_3(CH_2)_7CH=CH(CH_2)_7COOH$	*cis*-9-Octadecenoic acid	Oleic acid (Latin *oleum*, "oil")	13.4		
18:2($\Delta^{9,12}$)	$CH_3(CH_2)_4CH=CHCH_2CH=CH(CH_2)_7COOH$	*cis*-,*cis*-9,12-Octadecadienoic acid	Linoleic acid (Greek *linon*, "flax")	1–5		
18:3($\Delta^{9,12,15}$)	$CH_3CH_2CH=CHCH_2CH=CHCH_2CH=CH(CH_2)_7COOH$	*cis*-,*cis*-,*cis*-9,12,15-Octadecatrienoic acid	α-Linolenic acid	-11		
20:4($\Delta^{5,8,11,14}$)	$CH_3(CH_2)_4CH=CHCH_2CH=CHCH_2CH=CHCH_2CH=CH(CH_2)_3COOH$	*cis*-,*cis*-,*cis*-, *cis*-5,8,11,14-Icosatetraenoic acid	Arachidonic acid	–49.5		

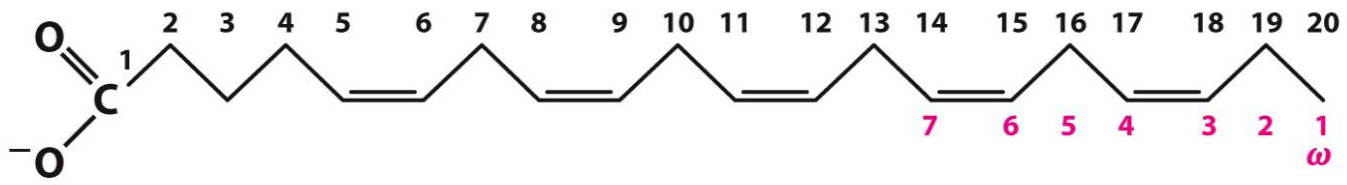
[a]All acids are shown in their nonionized form. At pH 7, all free fatty acids have an ionized carboxylate. Note that numbering of carbon atoms begins at the carboxyl carbon.
[b]The prefix *n*- indicates the "normal" unbranched structure. For instance, "dodecanoic" simply indicates 12 carbon atoms, which could be arranged in a variety of branched forms; "*n*-dodecanic" specifies the linear, unbranched form. For unsaturated fatty acids, the configuration of each double bond is indicated; in biological fatty acids the configuration is almost always cis.

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Essential fatty acids

- Two fatty acids are essential (we need them but our metabolism cannot synthesize them):
  - Linoleic acid: $18{:}2^{\Delta 9,12}$
  - α-linolenic acid (ALA): $18{:}3^{\Delta 9,12,15}$

- ALA is an ω-3 polyunsaturated fatty acid:
  - ω-3 PUFAs have been linked to protection against heart disease
  - Other ω-3 PUFAs can be synthesized from ALA: EPA, DHA



$20{:}5(\Delta^{5,8,11,14,17})$ Eicosapentaenoic acid (EPA)

Unnumbered 10 p363
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Solubility and melting point of fatty acids

- Solubility
  - decreases as the chain length increases

- Melting Point
  - decreases as the chain length decreases
  - decreases as the number of double bonds increases

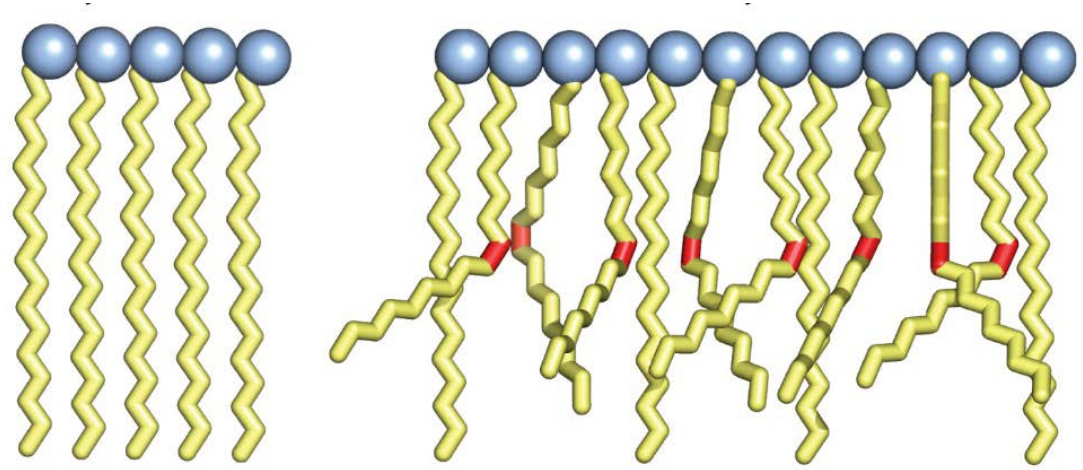


Figure 10-1cd
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Hydrogenation of fatty acids



The melting point of fatty acids can be controlled by the degree of hydrogenation



Watch this video about the molecular structure of margarine
https://www.youtube.com/watch?v=6J26Wmvltzs&feature=youtu.be

Trans fatty acids

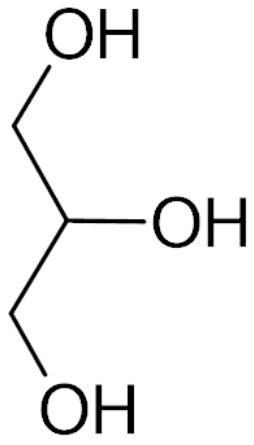
- *Trans* fatty acids form by partial hydrogenation of unsaturated fatty acids
  - done to increase stability at high temperature of oils used in cooking and to give foods a desirable taste and texture
- A *trans* double bond allows a given fatty acid to adopt an extended conformation
- *Trans* fatty acids can pack more regularly and show higher melting points than *cis* forms
- Intake of *trans* fats is associated with an increased risk of cardiovascular disease



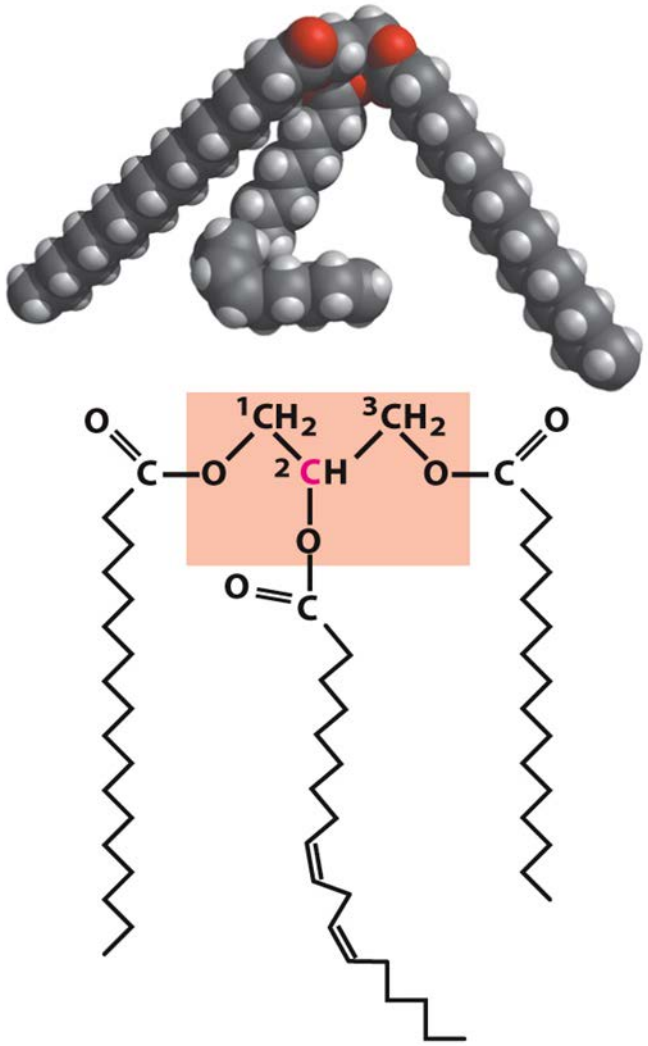
Triacylglycerols

- The majority of fatty acids in biological systems are found in the form of triacylglycerols
  - Solid ones are called fats
  - Liquid ones are called oils
- They are the primary storage form of lipids (body fat)
- More hydrophobic (less soluble in water) than fatty acids due to the esterification of the carboxylate group

Triacylglycerols



Glycerol



1-Stearoyl, 2-linoleoyl, 3-palmitoyl glycerol, a mixed triacylglycerol

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Fat vs. glycogen



How many calories do we store as fat in our body? And as glycogen?

In a regular person (70 kg):

125,000 kcal as fat

2,000 kcal as glycogen

Fats provide efficient fuel store

- The advantage of fats over polysaccharides:
  - Fatty acids carry more energy per carbon because they are more reduced
  - Fatty acids carry less water per gram because they are nonpolar
- Glucose and glycogen are for short-term energy needs: quick delivery
- Fats are for long-term energy needs: slow delivery

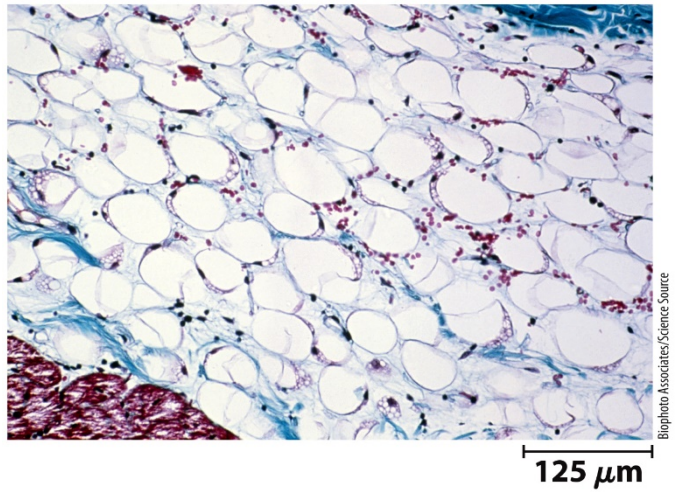


Figure 10-3a
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Biologically active lipids

Are present in much smaller amounts than storage or structural lipids:

- Steroid hormones
- Signaling molecules
- Lipid-soluble vitamins

Steroid hormones

- Steroids are derivatives of sterols
- Steroid hormones are synthesized from cholesterol in gonads and adrenal glands
- Many of the steroid hormones are male and female sex hormones

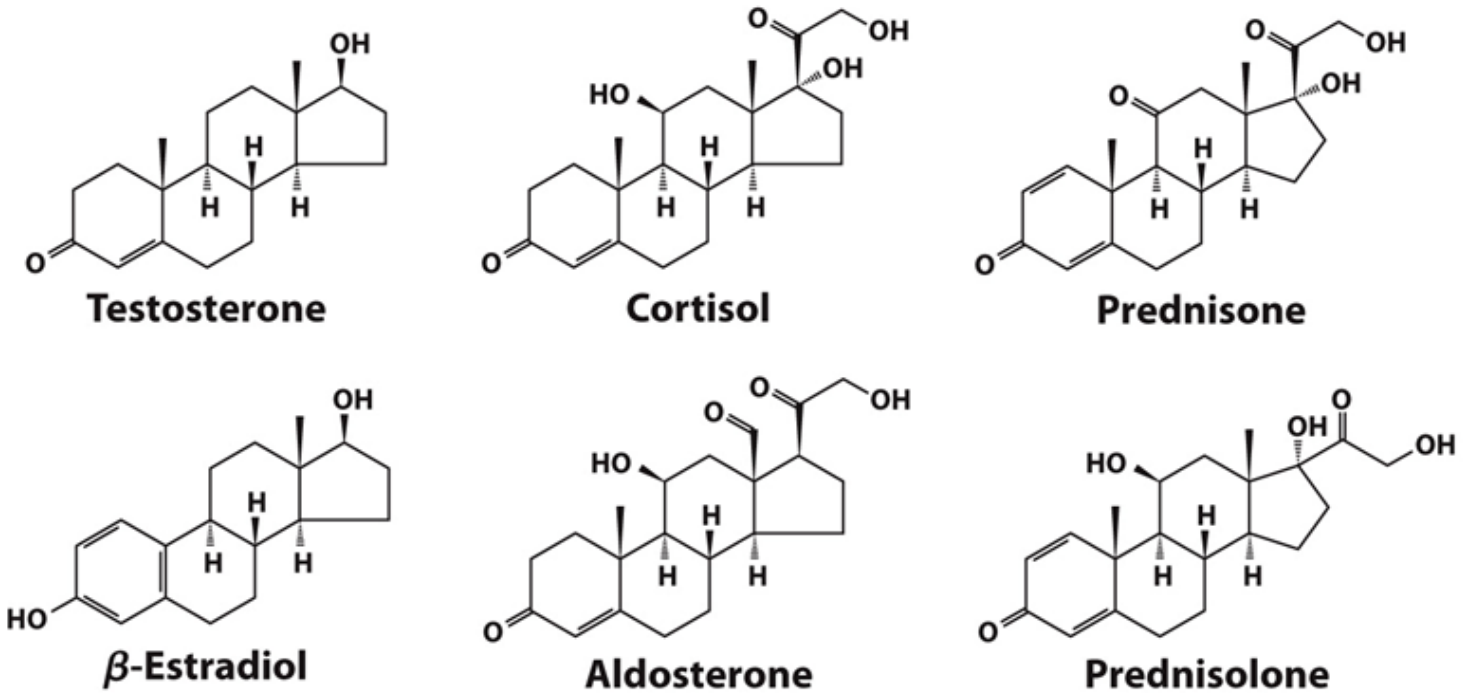


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Arachidonic acid derivatives

Enzymatic oxidation of arachidonic acid yields:

- prostaglandins (inflammation and fever)
- thromboxanes (formation of blood clots)
- leukotrienes (smooth muscle contraction in lungs)

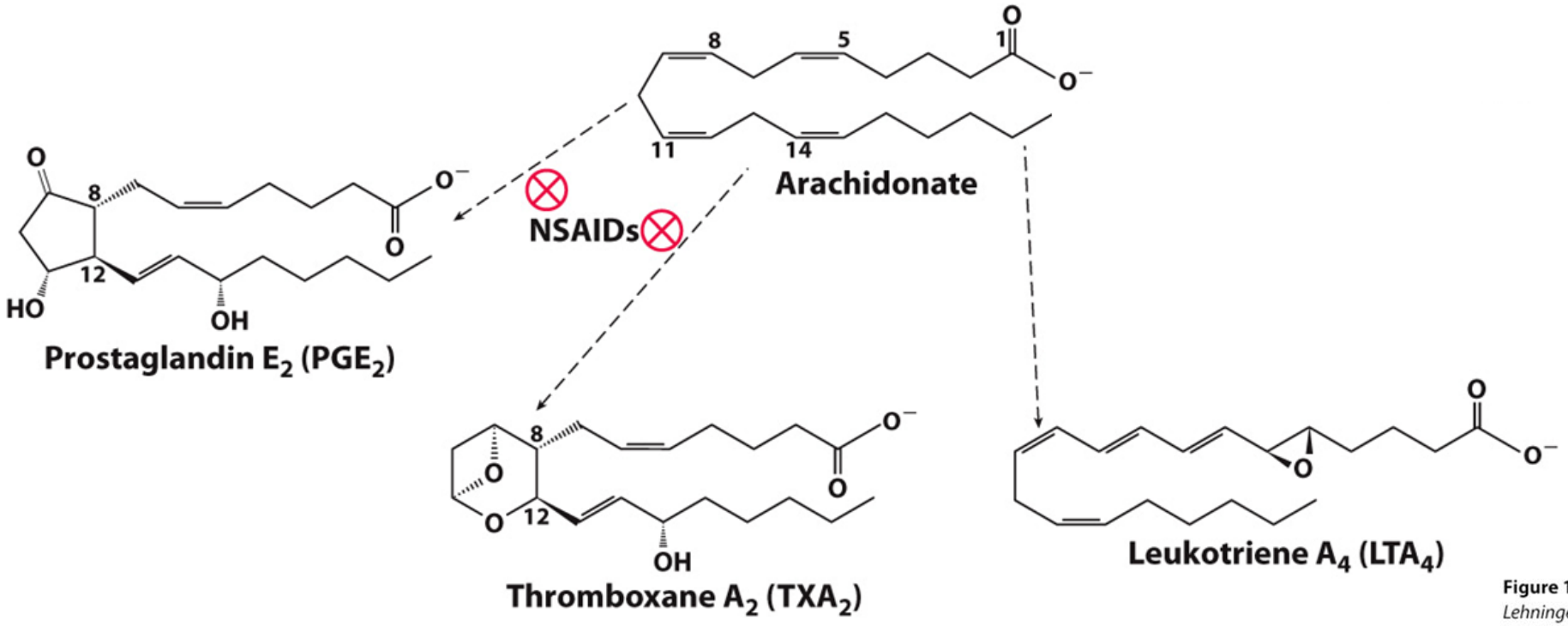
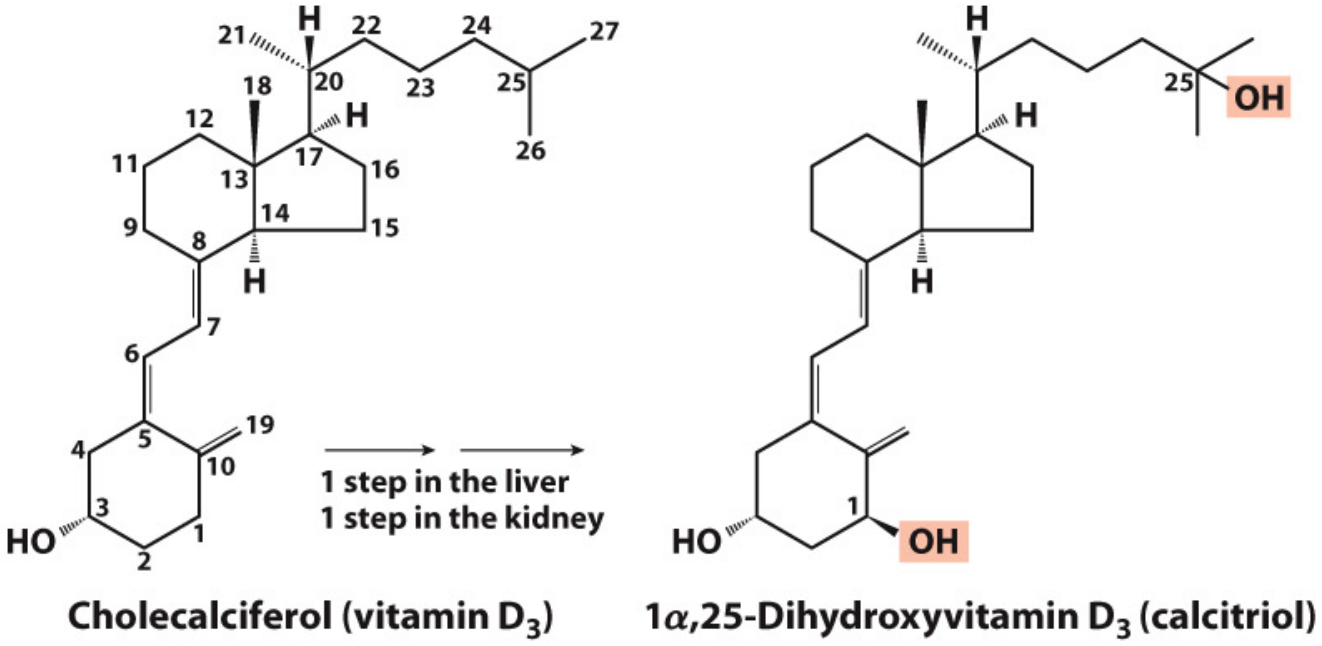


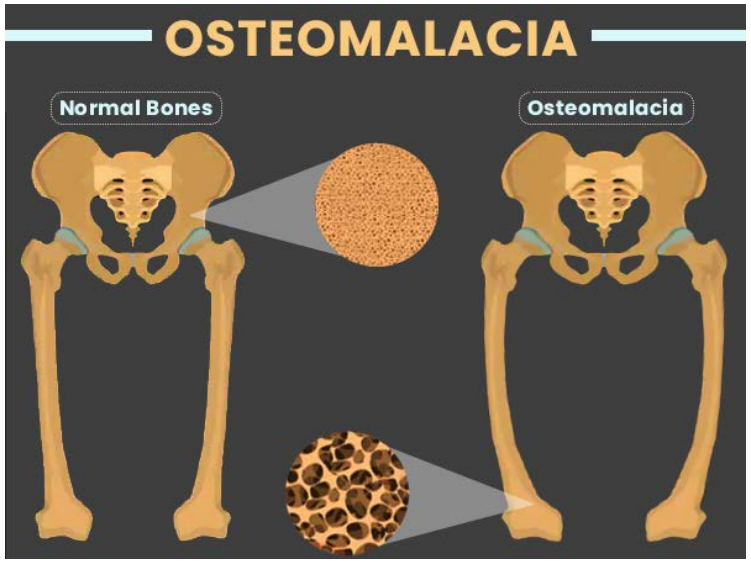
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Vitamin D regulates calcium metabolism



Cholecalciferol (vitamin D_3)

1α,25-Dihydroxyvitamin D_3 (calcitriol)

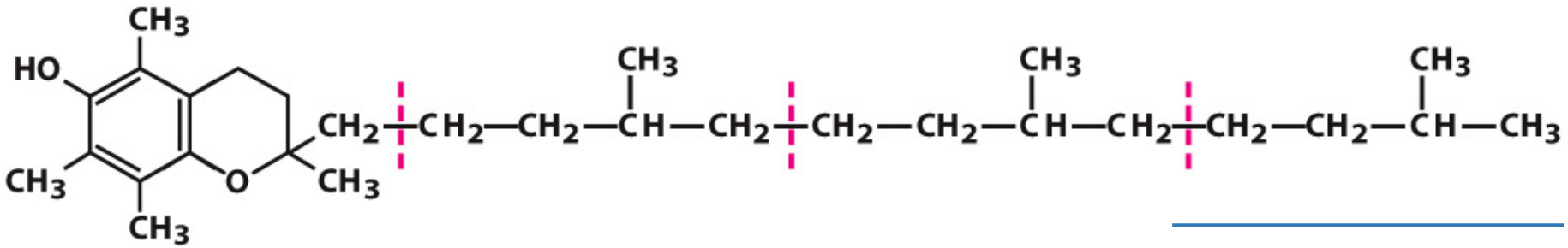


https://www.boldsky.com/health/

Rickets

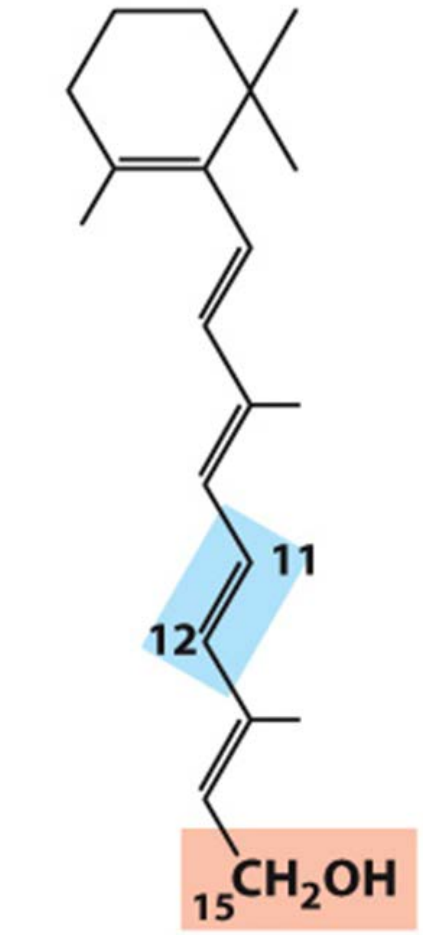


Vitamin E is antioxidant



Vitamin A (retinol)

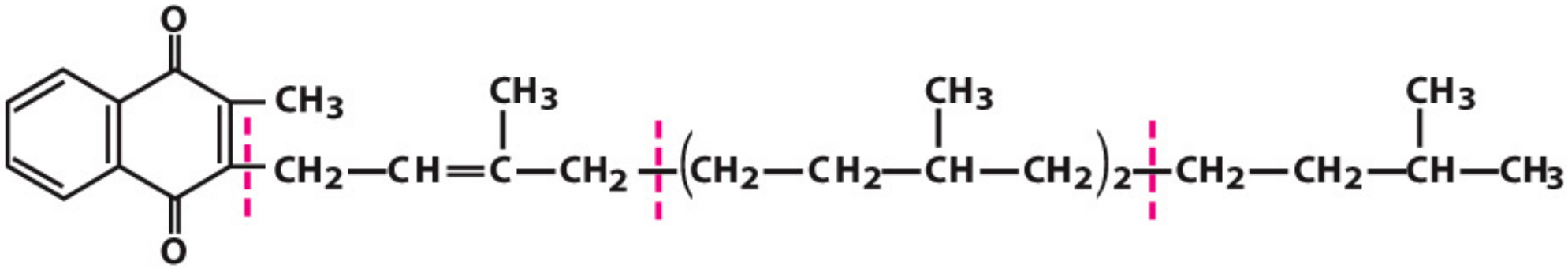
- Derived from β-carotene
- Involved in the visual cycle
- Precursor for other hormones involved in cell signaling



all-*trans*-Retinol (vitamin A_1)

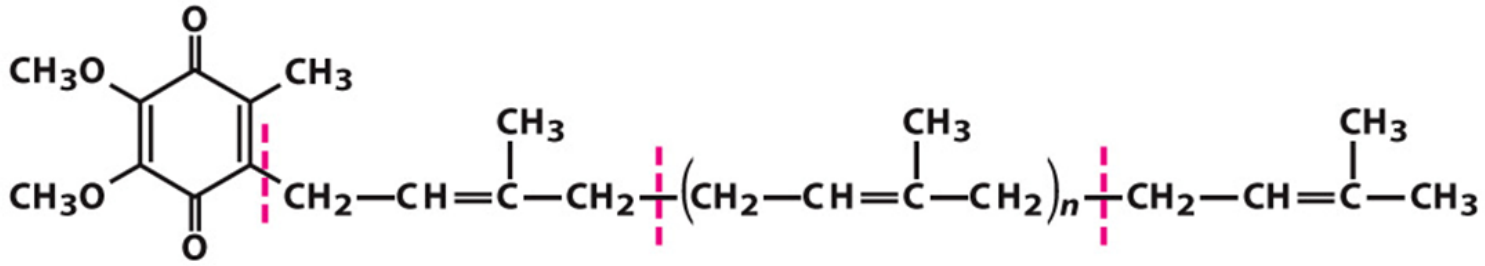
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Vitamin K is a clotting factor

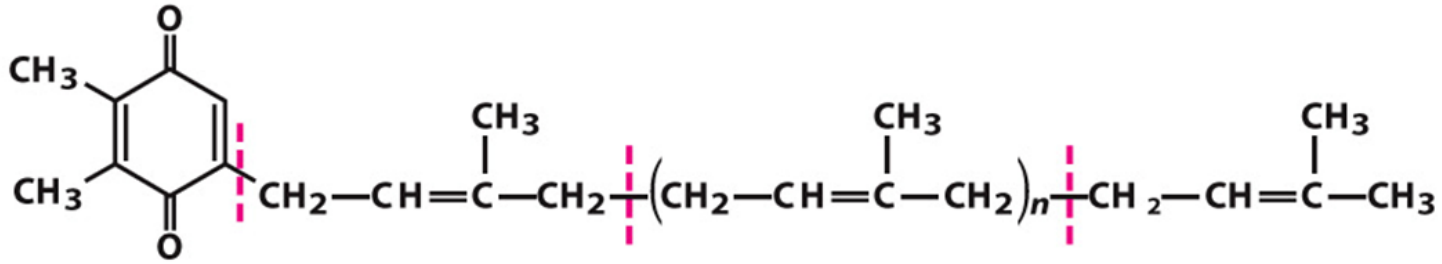


And many other isoprenoids …

Ubiquinone: a mitochondrial electron carrier (coenzyme Q)



Plastoquinone: a chloroplast electron carrier



Dolichol: a sugar carrier

